Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Disease Management

1.1 Cotton Leaf Curl Disease (CLCuD)

Disease Name and Causal Organism

- **Disease Name:** Cotton Leaf Curl Disease (CLCuD).
- **Causal Organism:** The disease is associated with viruses belonging to the genus *Begomovirus*, along with associated satellite DNA molecules.
- **Insect Vector:** The whitefly (*Bemisia tabaci*) transmits the virus complex.
- **Importance:** Cotton leaf curl disease is an important disease in cotton-growing areas of northern India, particularly Punjab, Haryana and Rajasthan.

Symptoms and Signs

- **Vein Thickening:** Small veins on young leaves become thickened and prominent.
- **Leaf Curling:** Leaves may curl upward or downward and become distorted.
- **Enations:** Abnormal leaf-like outgrowths, known as enations, may develop on the lower surface of leaves.
- **Leaf Deformation:** Infected leaves may become puckered, thickened or malformed.
- **Plant Stunting:** Internodes become shortened, resulting in reduced plant height and abnormal growth.
- **Reduced Reproductive Growth:** Severe infection can reduce flowering, square formation and boll development, resulting in yield losses.

Disease Development and Spread

- **Transmission:** The disease spreads primarily through whiteflies carrying the virus from infected plants to healthy cotton plants.
- **Infected Plants:** Infected cotton plants can serve as sources of infection for other plants.
- **Alternative Hosts:** Certain weeds and other host plants may support the virus or whitefly populations and contribute to disease spread.
- **Whitefly Movement:** Movement of whiteflies between infected plants and healthy cotton increases the risk of transmission.
- **Seasonal Risk:** The North Zone cotton guide identifies late June to August as an important period for transmission in that region. The timing may vary in other areas.

Conditions Favourable for Disease Development

- High populations of whiteflies in cotton fields.
- Presence of infected cotton plants and potential alternative host plants nearby.
- Weather conditions favourable for whitefly activity.
- Cultivation of susceptible cotton varieties or hybrids.
- Continued movement of whiteflies between infected and healthy plants.

Diagnosis and Identification

- Examine young leaves for thickening of small veins.
- Look for upward or downward leaf curling and leaf deformation.
- Check the lower leaf surface for characteristic enations.
- Observe whether internodes are shortened and plants are stunted.
- Assess flowering, square formation and boll development in affected plants.
- Check for whitefly activity as supporting evidence. Similar symptoms may have other causes, so laboratory confirmation may be required when the diagnosis is uncertain.

Prevention and Cultural Practices

- **Resistant Varieties:** Select locally recommended cotton varieties or hybrids with documented resistance or tolerance to cotton leaf curl disease.
- **Timely Sowing:** Follow the recommended sowing period for the local region to reduce exposure to high-risk periods where supported by regional guidance.
- **Regular Monitoring:** Inspect cotton fields frequently for early symptoms and whitefly activity.
- **Infected Plant Management:** Remove and dispose of infected plants where recommended by local agricultural authorities.
- **Weed Management:** Control weeds that are known alternative hosts of the virus or whitefly.
- **Field Sanitation:** Manage crop residues and volunteer cotton plants according to local recommendations.

Biological and Chemical Control

- **Biological Control:** Conserve naturally occurring whitefly predators and parasitoids where feasible as part of integrated pest management.
- **Chemical Control:** Where whitefly populations require treatment, use only locally registered products recommended by agricultural authorities and follow label directions.

- **Important Note:** Insecticides target the insect vector; they do not cure a plant already infected by the virus. Avoid unnecessary or repeated applications that may promote resistance or harm beneficial insects.

Treatment Methods and Application

Specific pesticide names, doses and application intervals should be included only after verification against current local recommendations and product registrations.

For the North Zone of India, consult the current recommendations from the relevant agricultural university or extension service before applying any regional treatment schedule in another area.

Integrated Disease Management

- Grow locally recommended resistant or tolerant varieties.
- Monitor whitefly populations and disease symptoms throughout the growing season.
- Follow regional sowing recommendations where they are supported by local disease-risk information.
- Manage infected plants and relevant alternative host plants according to local agricultural guidance.
- Use integrated whitefly management and conserve beneficial insects.
- Follow current pesticide registration, label directions and local agricultural recommendations.
- Combine preventive practices rather than relying on insecticide applications alone.

Monitoring and Early Detection

- Regularly inspect young leaves for vein thickening, leaf curling and enations.
- Observe whitefly populations, especially during locally identified high-risk periods.
- Mark suspected disease patches and monitor whether symptoms are spreading.
- Seek advice from an agricultural extension officer when symptoms are uncertain or disease incidence increases.

Effects on Crop Growth and Yield

Cotton leaf curl disease can cause severe stunting, shortened internodes, leaf deformation and reduced reproductive growth. Infected plants may produce fewer flowers and bolls, resulting in reduced yield. The extent of loss depends on the crop stage at infection, cultivar susceptibility and disease severity.

Other Relevant Information

- Cotton leaf curl disease is viral; fungicides do not directly cure the infection.
- Whitefly management can reduce transmission risk but cannot guarantee that plants will remain disease-free.

- Symptoms and disease severity vary with cultivar, region, season and whitefly pressure.
- Regional management recommendations should not be applied universally without checking local guidance.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, 2026. Cotton leaf curl disease and management sections.
2. *An Illustrative Guide for Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research, 2026. Cotton disease and cotton leaf curl disease sections.
3. *Integrated Pest Management Package for Cotton.* Government of India, 2014. Cotton disease management section.

1.2 Tobacco Streak Virus Disease (TSV)

Disease Name and Causal Organism

Tobacco Streak Virus Disease (TSV) is a viral disease of cotton caused by Tobacco streak virus (TSV), a member of the genus *Ilarvirus* in the family *Bromoviridae*. The disease has been reported in southern India and in parts of Maharashtra, including the Marathwada and Vidarbha regions.

Symptoms and Signs

- Early symptoms include chlorotic (yellowish) spots on young leaves.
- The spots may develop into purple, purplish-brown, or dark necrotic lesions.
- Veins, petioles, and leaf margins may show necrosis, while affected leaves can become distorted, puckered, or wrinkled.
- Dark necrotic streaks may appear along petioles, branches, and terminal shoots.
- Infected plants may show drying of squares (flower buds) and damage to terminal shoots.
- Severe infection can cause plant stunting, branch damage, and, when infection occurs early, premature plant death.
- Symptoms and severity may differ between cotton species and varieties.

Disease Development and Spread

TSV can survive in alternative host plants and weeds, particularly *Parthenium hysterophorus* (Congress grass). Transmission involves thrips and virus-bearing pollen grains. Thrips facilitate the movement of infected pollen to healthy cotton tissues during feeding, helping the virus spread within and between fields. The presence of Parthenium and increasing thrips populations can be associated with greater disease incidence.

Conditions Favourable for Disease Development

- Presence of Parthenium and other potential host weeds around fields, bunds, and irrigation channels.
- Increasing thrips populations and movement of virus-bearing pollen.
- Young cotton plants and susceptible varieties or hybrids.
- Field conditions that allow infected weeds and crop residues to remain between growing seasons.

Diagnosis and Identification

Inspect young leaves for chlorotic spots that develop into purplish-brown or dark necrotic lesions. Check for veinal necrosis, leaf deformation, dark streaks on petioles or branches, drying of squares, and terminal shoot damage. Also inspect field margins for Parthenium and check for thrips. Visual symptoms alone may not confirm TSV; laboratory diagnosis may be needed where confirmation is important. The provided sources do not specify a particular laboratory testing protocol.

Prevention and Cultural Practices

- Select resistant or tolerant varieties and hybrids recommended for the local cotton-growing region.
- Keep fields, bunds, irrigation channels, and nearby field margins free of Parthenium and other problematic weeds.
- Remove weeds before they flower and produce seed, following safe local weed-management practices.
- Inspect young cotton plants regularly and remove severely affected plants or plant parts where advised by local agricultural experts.
- Maintain field sanitation and remove cotton stalks after the final picking rather than allowing ratoon cotton to persist.
- Follow locally recommended crop rotation and intercropping practices where suitable.

Biological and Chemical Control

There is no direct curative spray that eliminates TSV from an infected cotton plant. Management focuses on reducing the risk of transmission and limiting sources of infection.

- Monitor thrips populations and use suitable monitoring methods, such as blue sticky traps, where recommended.
- Manage Parthenium and other relevant weeds to reduce potential reservoirs associated with the disease cycle.
- Use insecticides for thrips only when justified by local monitoring and official recommendations. Vector suppression does not guarantee prevention of viral infection.

- Before recommending any insecticide, confirm that the product is currently registered for the intended crop and pest, and follow its label, safety precautions, and pre-harvest interval.

Treatment Methods and Application

1. **Early scouting:** Begin regular inspection of young cotton plants and continue monitoring through vegetative growth, squaring, and flowering.
2. **Weed management:** Remove Parthenium and other target weeds in and around the field before they flower, using locally approved methods.
3. **Thrips monitoring:** Inspect leaves at different canopy levels and check blue sticky traps where these are part of the local monitoring programme.
4. **Infected plants:** Where recommended, remove severely affected plants or plant parts and dispose of them safely to reduce potential sources of infection.
5. **Chemical intervention:** If thrips levels reach the locally recommended action threshold, consult current agricultural extension advice for an approved product and application method. Do not rely on an insecticide to cure plants already infected with TSV.
6. **After harvest:** Remove cotton stalks and manage crop residues according to local recommendations to reduce the persistence of crop-related problems between seasons.

Important note: Specific pesticide products, doses, and economic thresholds vary with region, crop stage, and current registration. Confirm them with the relevant agricultural university or extension service before including them in farmer-facing advice.

Integrated Disease Management (IDM)

1. **Cultivar selection:** Choose locally recommended resistant or tolerant varieties and hybrids.
2. **Weed control:** Give particular attention to Parthenium around fields, bunds, irrigation channels, and nearby wasteland.
3. **Regular monitoring:** Inspect cotton for chlorotic and necrotic spots, veinal necrosis, leaf deformation, drying of squares, and terminal shoot injury.
4. **Thrips surveillance:** Monitor thrips populations using field inspections and suitable traps; act according to local thresholds and advisories.
5. **Sanitation:** Remove severely affected material where advised and clear cotton stalks after the final picking.
6. **Responsible pesticide use:** Use only currently approved products when needed, follow label directions, and avoid repeated use of the same mode of action.
7. **Local guidance:** Adapt the management plan to the local cotton-growing region and advice from agricultural experts.

Monitoring and Early Detection

Start scouting during the early growth stages and continue through squaring and flowering. Walk through the field in a consistent pattern, such as an X or Z, and inspect plants across different areas, paying attention to field edges, irrigation channels, and locations where weeds are present. Look for early chlorotic spots, purplish-brown necrosis, deformed leaves, veinal necrosis, and drying squares. Record the distribution of symptoms and observed thrips activity over time. Follow the monitoring frequency and sampling method recommended by local agricultural authorities.

Effects on Crop Growth and Yield

TSV can damage leaves, petioles, branches, terminal shoots, and reproductive structures. Drying of squares and severe stunting can reduce the plant's ability to form and retain bolls. Infection at an early stage may cause premature plant death. The ICAR–CICR guide reports that yield loss can be substantial under severe infection, although actual losses depend on disease severity, variety, crop stage, and local conditions.

Other Relevant Information

- TSV is a viral disease; fungicides are not a direct cure.
- Thrips, virus-bearing pollen, and alternative host plants are important parts of the disease cycle described in the ICAR–CICR guides.
- Parthenium management is an important preventive measure in the cited guidance.
- Symptoms may resemble other diseases or physiological disorders. Seek expert or laboratory confirmation when field diagnosis is uncertain.
- Management recommendations from the North Zone guide should not automatically be applied to Maharashtra or other regions without local validation.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, Regional Station, Sirsa, 2026. Section on Tobacco Streak Virus (TSV).
2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Section on Tobacco Streak Virus Disease, including disease development, surveillance, and management.

1.3 Bacterial Leaf Blight of Cotton

Disease Name and Causal Organism

- **Disease Name:** Bacterial Leaf Blight of Cotton (BLB), also associated with blackarm symptoms and boll infection.
- **Causal Organism:** The disease is caused by the bacterium *Xanthomonas citri* pv. *malvacearum.*

- **Importance:** Bacterial leaf blight can affect cotton at different growth stages and may cause leaf damage, defoliation, damage to stems and branches, and deterioration of yield and fibre quality.

Symptoms and Signs

- **Water-Soaked Lesions:** Initial symptoms appear as small, water-soaked spots on leaves of seedlings and mature plants.
- **Angular Leaf Spots:** The lesions enlarge and develop angular shapes because their spread is restricted by leaf veins.
- **Oily Appearance:** The lower surface of affected leaves may show a characteristic oily or greasy appearance.
- **Yellow Halo:** Lesions on the upper leaf surface may be surrounded by a yellow halo.
- **Dark Lesions:** Older lesions become darker, often black, and may enlarge, resulting in premature leaf fall.
- **Blackarm Symptoms:** Infection involving leaf veins and extending into stems or branches may produce dark lesions commonly associated with the blackarm symptom.
- **Boll Damage:** Infection can also affect cotton bolls, potentially reducing seed and fibre quality.

Disease Development and Spread

- **Seed Transmission:** The bacterium can be carried on or in contaminated seed, which may introduce the pathogen into a new crop.
- **Crop Residues:** The pathogen may survive in infected, dried plant material and crop debris.
- **Rain Splash:** Wind-driven rain and splashing water can transfer bacteria from infected material to healthy plant parts.
- **Wind and Irrigation:** Wind-driven rain and sprinkler irrigation may contribute to the spread of the bacterium.
- **Farm Operations:** Contaminated tools and other field operations may contribute to disease spread.
- **Secondary Infection:** Under favourable conditions, bacteria can spread from existing infection sites to healthy tissues and neighbouring plants.

Conditions Favourable for Disease Development

- High humidity and frequent rainfall.
- Rain splash and wet foliage that favour bacterial movement.
- Windy conditions during periods of rainfall.
- Use of contaminated seed or presence of infected crop residues.

- Dense crop growth and field conditions that allow infected foliage to remain wet for extended periods.

Diagnosis and Identification

- Examine leaves for small, water-soaked lesions during early stages of infection.
- Look for angular spots whose margins are limited by leaf veins.
- Check the lower leaf surface for an oily or greasy appearance.
- Observe whether lesions have yellow halos or develop into dark or black spots.
- Inspect veins, petioles, stems and branches for dark lesions associated with blackarm symptoms.
- Examine affected bolls for signs of infection and deterioration.
- Similar symptoms may occur with other diseases. Laboratory confirmation may be required when the diagnosis is uncertain.

Prevention and Cultural Practices

- **Resistant Varieties:** Select resistant or tolerant cotton varieties and hybrids recommended for the local region.
- **Healthy Seed:** Use certified, healthy seed from reliable sources to reduce the risk of introducing the pathogen.
- **Seed Treatment:** Use only seed-treatment products recommended by local agricultural authorities and approved for the intended use.
- **Field Sanitation:** Remove and manage infected crop residues after harvest according to local recommendations.
- **Water Management:** Where practical, avoid irrigation practices that unnecessarily wet foliage, particularly when disease is present.
- **Tool Hygiene:** Clean farm tools and equipment when moving between affected and healthy areas, following appropriate sanitation practices.
- **Regular Monitoring:** Inspect cotton plants throughout the season for early symptoms and signs of disease spread.

Biological and Chemical Control

- **Biological Control:** Use biological seed-treatment products only where they are recommended by a recognised agricultural authority for this disease and crop.
- **Chemical Control:** The ICAR–CICR North Zone guide recommends seed treatment and need-based chemical interventions as part of disease management. Select products only after confirming current registration and local recommendations.
- **Preventive Approach:** Healthy seed, resistant varieties, sanitation and early monitoring are important components of management.

- **Important Note:** Bacterial leaf blight is caused by a bacterium, not a fungus. Fungicides should not be assumed to control it effectively. Product choice must be based on approved recommendations for bacterial disease management.

Treatment Methods and Application

Specific pesticide names, doses, seed-treatment rates and spray intervals should be included only after verification against current local recommendations and product registrations.

1. **Before Sowing:** Obtain healthy seed and follow locally approved seed-treatment recommendations.
2. **Early Crop Inspection:** Inspect seedlings and developing plants for water-soaked and angular leaf lesions.
3. **Field Sanitation:** Remove and manage infected crop residues according to local agricultural guidance.
4. **During Wet Weather:** Increase field inspections when rainfall, wind-driven rain or prolonged leaf wetness may favour disease development.
5. **Chemical Intervention:** If treatment is advised, use only a currently registered product recommended for the crop and disease, following its label directions.
6. **After Harvest:** Remove or manage infected stalks and plant debris according to local recommendations to reduce potential sources of infection for the next crop.

Integrated Disease Management

- Grow resistant or tolerant cotton varieties and hybrids recommended for the local region.
- Use healthy seed and follow approved seed-treatment recommendations.
- Inspect the crop regularly for water-soaked spots, angular lesions and dark vein or stem symptoms.
- Manage infected plant debris and maintain field sanitation.
- Reduce unnecessary foliage wetting where practical and use irrigation practices appropriate for the local conditions.
- Clean contaminated tools and equipment where appropriate.
- Use chemical interventions only when supported by current local recommendations and approved product labels.
- Combine preventive measures rather than relying solely on chemical applications.

Monitoring and Early Detection

- Inspect seedlings and mature plants for early water-soaked lesions.
- Check whether spots are angular and restricted by veins.
- Examine the underside of leaves for an oily or greasy appearance and the upper surface for yellow halos.
- Monitor for darkening lesions, premature defoliation, and symptoms on veins, stems and branches.
- Increase monitoring during periods of high humidity, rainfall and wind-driven rain.
- Record the affected areas and seek advice from an agricultural extension officer if disease incidence increases or diagnosis is uncertain.

Effects on Crop Growth and Yield

Bacterial leaf blight can damage leaves, veins, stems, branches and bolls. Severe leaf infection may cause premature defoliation and reduce the plant's capacity to support normal growth and boll development. Boll infection can affect seed and fibre quality. The extent of yield and quality loss depends on disease severity, crop stage, variety and environmental conditions.

Other Relevant Information

- Bacterial leaf blight may also be referred to as blackarm when characteristic dark symptoms occur on stems or branches.
- The ICAR–CICR North Zone guide identifies contaminated seed, infected plant residues and rain-associated spread as important parts of the disease cycle.
- Disease severity can increase under humid conditions and when rainfall or irrigation splashes move bacteria between plant tissues.
- Leaf symptoms can resemble those of other diseases; diagnosis should consider the full pattern of symptoms and, where necessary, laboratory testing.
- Management recommendations should be checked with the relevant agricultural university or extension service before being applied outside the region for which they were developed.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, 2026. Bacterial leaf blight symptoms, disease cycle and management sections.
2. *An Illustrative Guide for Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research, 2026. Cotton disease identification and management sections.
3. *Integrated Pest Management Package for Cotton.* Government of India, 2014. Cotton disease management section.

1.4 Fungal Root Rot Disease of Cotton

Disease Name and Causal Organisms

- **Disease Name:** Fungal Root Rot of Cotton.
- **Causal Organisms:** Cotton root rot is a disease complex primarily associated with the soil-borne fungi *Rhizoctonia solani*, *Rhizoctonia bataticola* (also known as *Macrophomina phaseolina*), and *Sclerotium rolfsii*.
- **Nature of the Disease:** The disease affects the roots and collar region, reducing the plant's ability to absorb and transport water and nutrients. Severely affected plants may wilt suddenly, become easy to uproot, and eventually die.
- **Primary Sources of Infection:** The pathogens can survive in soil, infected crop residues, and contaminated seed material. The relative contribution of each pathogen depends on the field and environmental conditions.

Symptoms and Identification

- **Sudden Wilting:** Infected plants may show sudden wilting and drooping, beginning at the upper part of the plant and progressing downward.
- **Easy Uprooting:** Affected plants can often be pulled out of the soil easily because the roots have decayed and lost their strength.
- **Root Bark Disintegration:** The outer root tissues may disintegrate into shreds, sometimes giving the roots a yellowish appearance.
- **Symptoms Caused by *Rhizoctonia solani*:** Roots may become brown, wet, and soft.
- **Symptoms Caused by *Rhizoctonia bataticola*:** Roots may turn black, dry, and brittle.
- **Symptoms Caused by *Sclerotium rolfsii*:** Profuse white fungal growth may appear around the roots and collar region. Small, spherical structures called sclerotia may also develop, followed by root decay and seedling death.
- **Patchy Field Appearance:** Diseased plants may occur in patches, so the crop should be inspected systematically rather than checking only individual wilted plants.
- **Important Diagnostic Note:** Wilting can also result from other root diseases, water stress, and vascular wilt pathogens. Uproot a few symptomatic plants carefully and inspect the roots and collar region before deciding on treatment.

Disease Rating Scale

The ICAR–CICR guide provides the following field-incidence scale for root rot:

Disease Rating Scale

- **Scale 0:** Plants remain completely free from infection.
- **Scale 1:** Approximately 5% disease incidence.

- **Scale 2:** Approximately 6–15% disease incidence.
- **Scale 3:** Approximately 16–25% disease incidence.
- **Scale 4:** More than 26% disease incidence.

Note: Record the percentage of affected plants across representative parts of the field. Use the same sampling approach during successive inspections to assess whether the disease is increasing.

Disease Cycle and Epidemiology

- **Survival in Soil and Crop Residues:** Root rot pathogens can survive for several years in soil and infected plant residues. Some may also persist on the fuzzy material of contaminated seeds.
- **Primary Infection:** Infection may begin when susceptible roots come into contact with viable pathogens present in the soil or associated with contaminated seed material.
- **Secondary Spread:** Movement of contaminated soil, irrigation water, rain splash, and farm tools can contribute to the spread of inoculum within a field.
- **Favourable Conditions:** Disease severity can be high in subhumid and semiarid environments, particularly when plants experience dry conditions or moisture stress. Soil conditions and the specific pathogen involved influence disease development.
- **Crop Stress:** Irregular irrigation, root injury, and other stresses may worsen plant damage and make disease management more difficult.
- **Persistence Between Seasons:** Infected roots and undecomposed crop residues may maintain the pathogens in the field after harvest, increasing the importance of sanitation and residue management.

Favourable Conditions for Disease Development

- Presence of root rot pathogens in soil, crop residues, or contaminated seed material.
- Dry spells and moisture stress, which can increase the risk or severity of root rot in susceptible cotton crops.
- Poorly managed irrigation, particularly irregular moisture supply that places plants under stress.
- Root injury during cultivation or other field operations, which can further compromise plant health.
- Continuous cultivation of susceptible crops in fields with a history of root rot, without effective sanitation and other preventive measures.

Prevention and Cultural Management

- **Field Monitoring:** Inspect the crop regularly, especially during early growth and periods of moisture stress. Mark affected patches so that they can be checked during subsequent visits.

- **Removal of Infected Plants:** Promptly remove and destroy severely affected plants, including infected roots where practical, in accordance with local agricultural guidance. Avoid moving contaminated soil to healthy areas.
- **Crop Residue Management:** Manage infected cotton residues after harvest to reduce the amount of potentially infected material remaining in the field.
- **Balanced Fertilizer Use:** Avoid excessive nitrogen application. Apply nutrients according to soil-test results and locally recommended fertilizer schedules.
- **Moisture Management:** Maintain an appropriate irrigation schedule to avoid moisture stress. Prevent prolonged waterlogging and avoid abrupt, poorly managed changes in soil moisture.
- **Careful Field Operations:** Minimize root damage during intercultural operations and clean tools when moving between affected and healthy field areas.
- **Variety Selection:** Prefer locally recommended varieties or hybrids with suitable disease tolerance where reliable recommendations are available. Do not assume that a variety is resistant unless this has been established for the relevant region and pathogen.

Biological Management

- **Use of *Trichoderma*:** Suitable formulations of beneficial fungi such as *Trichoderma harzianum* or *Trichoderma viride* may be used as part of an integrated root rot management programme.
- **Root-Zone Application:** The ICAR–CICR integrated management guide describes root-zone drenching with a *Trichoderma* formulation when root rot is detected. The formulation, concentration, application method, and timing must follow the current product label and local agricultural recommendations.
- **Preventive Use:** Where recommended, apply biological products preventively or at the crop stage specified by the product label. Results may vary with soil conditions, pathogen pressure, product quality, and application method.
- **Compatibility:** Avoid mixing biological products with chemical fungicides unless compatibility is specifically confirmed on the label or by a qualified agricultural adviser.

Chemical Management

- **Need-Based Treatment:** Chemical treatment should be considered only after the disease has been reasonably identified and a locally approved product is available for cotton root rot.
- **Root-Zone Drenching:** The ICAR–CICR integrated management guide lists carbendazim 50% WP as a root-zone drenching option. Use it only if currently registered and labelled for this purpose in the relevant location, and follow the approved label rate, precautions, and application instructions.

- **Avoid Unverified Recommendations:** Do not apply a fungicide solely on the basis of a general symptom such as wilting. Different root pathogens and non-infectious causes may require different management approaches.
- **Safe Application:** Use the required personal protective equipment, observe label restrictions and re-entry instructions, and prevent contamination of water sources. Do not increase the dose or frequency beyond approved directions.
- **Local Validation:** Confirm current product registration and recommendations with the local agricultural extension service or Krishi Vigyan Kendra before using a chemical treatment.

Integrated Disease Management (IDM)

1. Use healthy, quality-assured seed and follow locally recommended seed-treatment practices.
2. Monitor fields regularly for sudden wilting, root discolouration, root decay, and white fungal growth near the collar region.
3. Inspect symptomatic plants by carefully examining the roots and collar region to help distinguish root rot from other causes of wilting.
4. Remove and destroy infected plants promptly where practical, while avoiding the movement of contaminated soil and plant material into healthy areas.
5. Maintain balanced nutrition and a suitable irrigation schedule to reduce crop stress.
6. Manage infected crop residues after harvest and maintain field sanitation.
7. Use a suitable biological product such as *Trichoderma* where recommended and compatible with the crop protection programme.
8. If chemical treatment is justified, use only a currently approved product according to its label and local recommendations.
9. Record disease incidence, affected field patches, crop stage, irrigation conditions, and management actions to support decisions in the current and following seasons.

Monitoring and Field-Level Decision Making

- **What to Monitor:** Sudden wilting, drooping, easy uprooting, root bark disintegration, black and brittle roots, brown and soft roots, and white fungal growth around the roots or collar.
- **Where to Inspect:** Check both affected patches and apparently healthy plants nearby. Examine the distribution of symptoms across the field.
- **When to Act:** Investigate promptly when new wilted plants appear or existing affected patches expand. Early identification helps prevent avoidable crop loss and unnecessary pesticide applications.
- **Records:** Note the date, crop growth stage, approximate percentage of affected plants, recent irrigation and rainfall, root symptoms, and action taken.

- **Seek Confirmation:** If symptoms are unclear, rapidly increasing, or inconsistent with typical root rot symptoms, seek diagnosis from an agricultural extension officer or plant pathology laboratory before selecting a treatment.

Effects on Crop Growth and Yield

- Damaged roots reduce the plant's capacity to absorb water and nutrients.
- Infected plants may wilt, lose vigour, and die, resulting in gaps in the crop stand.
- Severe disease can reduce the number of productive plants and consequently lower yield.
- The extent of yield loss depends on disease incidence, the pathogens present, crop stage, cotton variety or hybrid, and environmental conditions. A single yield-loss percentage should not be assumed for every field.

Important Information for Farmers

- Root rot is a complex disease involving more than one soil-borne fungal pathogen. Root symptoms can help with preliminary identification, but laboratory confirmation may be necessary.
- Sudden wilting alone does not confirm root rot. Inspect the root system before deciding on disease-specific treatment.
- Field sanitation, appropriate irrigation, balanced fertilizer use, and early removal of infected plants are important components of management.
- Biological and chemical treatments should be selected according to current local recommendations. Their effectiveness depends on correct diagnosis, formulation quality, timing, and application.
- Recommendations in the cited North Zone guide are region-specific. Farmers in other cotton-growing regions should confirm suitability with their local agricultural extension service.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, 2026. Fungal root rot disease identification, symptoms, disease cycle, and management sections, pp. 107–108.
2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Root rot management recommendations, including moisture management, sanitation, biological management, and root-zone treatment.

1.5 Fusarium Wilt Disease of Cotton

Disease Name and Causal Organism

- **Disease Name:** Fusarium Wilt of Cotton.

- **Causal Organism:** The disease is caused by the soil-borne fungus *Fusarium oxysporum* f. sp. *vasinfectum*.
- **Affected Cotton Species:** The disease predominantly affects Desi cotton (*Gossypium arboreum*) and Asiatic cotton (*Gossypium herbaceum*). The ICAR–CICR guide also reports that certain races have recently been associated with wilt in upland cotton.
- **Typical Occurrence:** The disease commonly appears between 30 and 120 days after sowing, although infection can occur from the seedling stage through boll development.

Symptoms and Identification

- **Seedling Symptoms:** Initial symptoms may appear as necrotic lesions on the cotyledons. These lesions can extend to newly developing leaves.
- **Leaf Symptoms:** Reticulate or net-like necrosis may develop along the leaf veins. Affected leaves may droop and gradually wilt.
- **Wilting and Drying:** As the disease progresses, infected plants may show partial or complete drying. Symptoms can occur during vegetative growth and boll development.
- **Vascular Discolouration:** The vascular tissues in roots, stems, and petioles may show dark-brown discolouration. This can be observed by cutting the affected tissue longitudinally.
- **Plant Growth:** Infected plants may lose vigour, wilt, and eventually die, depending on disease severity and crop conditions.
- **Role of Nematodes:** Nematode infestation may promote infection and increase disease severity at different stages of crop growth.
- **Diagnostic Note:** Wilting and vascular discolouration can also occur in other diseases. Examine the symptoms carefully and seek confirmation from an agricultural specialist when the cause is uncertain.

Disease Rating Scale

The ICAR–CICR guide provides the following scale based on disease incidence:

- **Scale 0:** All plants in the entry or field remain free from infection.
- **Scale 1:** Approximately 5% disease incidence.
- **Scale 2:** Approximately 6–15% disease incidence.
- **Scale 3:** Approximately 16–25% disease incidence.
- **Scale 4:** More than 26% disease incidence.

Note: Inspect representative areas of the field and record the percentage of affected plants. Use a consistent sampling method for comparisons during subsequent inspections.

Disease Cycle and Epidemiology

- **Seed-Borne and Soil-Borne Survival:** The pathogen can be carried by seed and survive in soil for several years, including in association with contaminated, dried plant residues.
- **Primary Infection:** Infected seed material and infested soil can act as sources of primary infection.
- **Secondary Spread:** The pathogen may spread within the field through irrigation water, rain splash, and contaminated farm tools.
- **Persistence in Soil:** Long-term survival in soil and plant residues can allow the disease to reappear in subsequent cotton seasons.
- **Environmental Conditions:** The disease is reported to be severe in subhumid and semiarid regions. Heavy black and alkaline soils are also identified as favourable conditions in the ICAR–CICR guide.
- **Pathogen Variability:** Different physiological races of *Fusarium oxysporum* f. sp. *vasinfectum* have been reported, and their ability to affect cotton may differ.

Favourable Conditions for Disease Development

- Presence of the pathogen in soil, infected plant residues, or contaminated seed material.
- Growing susceptible cotton varieties or hybrids in fields with a history of Fusarium wilt.
- Conditions associated with heavy black and alkaline soils, as reported in the ICAR–CICR North Zone guide.
- Root damage or nematode infestation that may contribute to infection or increase disease severity.
- Repeated cotton cultivation without appropriate field sanitation or other preventive measures.

Prevention and Cultural Management

- **Resistant Varieties:** Grow varieties or hybrids recommended for the local region, giving preference to those with documented resistance or tolerance to Fusarium wilt where available.
- **Healthy Seed:** Use quality-assured seed from reliable sources and follow locally recommended seed-treatment practices.
- **Crop Rotation:** Adopt a suitable crop rotation based on local agricultural recommendations and the field's disease history. Rotation should be planned carefully because the pathogen can persist in soil and may have a broad host range.
- **Field Sanitation:** Remove and manage infected plant residues appropriately to reduce the persistence of disease sources.

- **Tool Hygiene:** Clean soil and plant material from farm tools when moving between affected and healthy areas to reduce the risk of transferring contaminated material.
- **Nematode Monitoring:** Where plants show root damage or the field has a history of nematode problems, seek appropriate diagnosis and management advice because nematode infestation may increase disease severity.
- **Field History:** Maintain records of affected patches, previous crops, disease incidence, and management actions to guide future crop planning.

Biological Management

- **Recommended Biopesticides:** The ICAR–CICR guide recommends seed and soil treatment using suitable biopesticides as part of disease management.
- **Product Selection:** Use only biological formulations approved or recommended for the intended application and follow the product label for dosage, timing, and method of application.
- **Preventive Approach:** Biological treatments should be combined with healthy seed, suitable varieties, crop rotation, field sanitation, and regular monitoring rather than relied upon as the sole control measure.
- **Compatibility:** Do not mix biological products with chemical fungicides unless product instructions or a qualified agricultural adviser confirm compatibility.

Chemical Management

- **Seed and Soil Treatment:** The ICAR–CICR guide recommends seed and soil treatment with locally recommended chemicals and biopesticides.
- **Fungicide Use:** Fungicides should be selected only when they are currently approved for the intended use in cotton and are recommended for the diagnosed disease in the relevant region.
- **Correct Application:** Follow the approved label for product rate, application method, crop stage, protective equipment, and other safety precautions. Do not increase the dose or frequency beyond the label directions.
- **Avoid Unnecessary Sprays:** Because Fusarium wilt is associated with infection in soil and vascular tissues, do not assume that a routine foliar spray will control the disease.
- **Local Advice:** Confirm the current treatment recommendation with the local agricultural extension service or Krishi Vigyan Kendra before using a chemical product.

Integrated Disease Management (IDM)

1. Use quality-assured seed and follow locally recommended seed-treatment practices.
2. Select varieties or hybrids recommended for the region, with documented resistance or tolerance to Fusarium wilt where available.

3. Monitor the crop from the seedling stage through boll development for necrotic leaf lesions, drooping, wilting, and plant drying.
4. Inspect affected plants for dark-brown vascular discolouration in the roots, stems, and petioles.
5. Keep records of previous disease occurrence and use crop rotation where suitable for the local cropping system.
6. Manage infected plant residues and clean farm tools to reduce the movement of contaminated material.
7. Investigate possible nematode infestation where root damage or a field history suggests that nematodes may be contributing to disease severity.
8. Use recommended biological or chemical seed and soil treatments only according to current local guidance and product-label instructions.
9. Record disease incidence and the locations of affected patches to help evaluate management and plan for the next season.

Monitoring and Field-Level Decision Making

- **What to Monitor:** Necrotic cotyledon lesions, net-like necrosis along leaf veins, leaf drooping, wilting, plant drying, and dark-brown vascular discolouration.
- **When to Monitor:** Inspect the crop regularly from the seedling stage until boll development, with particular attention to fields with a history of wilt.
- **Where to Inspect:** Examine plants across representative parts of the field, including patches where wilting or poor growth is observed.
- **Disease Assessment:** Estimate the percentage of infected plants using the disease rating scale and record changes over time.
- **Seek Confirmation:** If symptoms are unclear or disease incidence increases rapidly, consult an agricultural specialist or plant pathology laboratory before choosing a treatment.

Effects on Crop Growth and Yield

- Vascular damage interferes with the movement of water and nutrients through the plant.
- Infected plants may lose vigour, wilt, dry partially or completely, and die.
- Severe infection can reduce the number of productive plants and affect crop establishment and yield.
- The actual yield loss depends on disease incidence, cotton variety or hybrid, pathogen race, crop stage, soil conditions, and the effectiveness of management practices.

Important Information for Farmers

- Fusarium wilt can occur at different stages of cotton growth, from seedlings through boll development.
- Wilting alone is not sufficient to confirm Fusarium wilt. Inspect vascular tissues and consider other possible causes of plant wilting.
- The pathogen can persist in soil and plant residues for several years, making preventive management and field history important.
- Nematode infestation may increase disease severity and should be considered when investigating affected fields.
- Management should combine recommended varieties, healthy seed, suitable seed and soil treatment, crop rotation, sanitation, and regular monitoring.
- Recommendations in the cited ICAR–CICR guide are specific to its stated production context. Confirm the suitability of varieties and treatments with local agricultural authorities before implementation.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, 2026. Fusarium wilt disease identification, symptoms, disease cycle, rating scale, and management section, pp. 110–111.
2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Integrated disease management guidance for cotton.

1.6 Target Leaf Spot Disease of Cotton

Disease Name and Causal Organism

- **Disease Name:** Target Leaf Spot of Cotton.
- **Causal Organism:** The disease is caused by the fungal pathogen *Corynespora cassiicola*.
- **Nature of the Disease:** Target leaf spot is a fungal foliar disease that affects cotton leaves and can reduce healthy leaf area when infection becomes severe.
- **Host Range:** The pathogen has a wide host range, so disease management should include field sanitation and monitoring of infected crop material.

Symptoms and Identification

- **Initial Symptoms:** Small, circular leaf spots may develop on the leaves.
- **Target-Shaped Lesions:** The characteristic symptom is the development of concentric rings on the lesions, giving them a target-like appearance. A distinct dark spot may be visible near the centre.
- **Lesion Development:** As the disease progresses, spots may enlarge and merge, causing larger areas of damaged leaf tissue.

- **Leaf Damage:** Severely affected leaves may develop extensive necrotic areas and dry prematurely.
- **Premature Defoliation:** Under favourable conditions, infection may cause premature leaf fall, reducing the functional leaf area of the plant.
- **Diagnostic Note:** Several cotton diseases and non-infectious problems can cause leaf spots. Target-like concentric rings are an important identification feature, but uncertain cases should be checked by an agricultural specialist.

Disease Rating and Field Assessment

- **Field Monitoring:** Inspect representative plants in different parts of the field and record the presence and spread of target-shaped leaf lesions.
- **Early Disease:** Record isolated lesions and note whether they are concentrated on particular leaves or parts of the canopy.
- **Moderate Disease:** Monitor whether individual spots are enlarging or multiple lesions are appearing on the same leaves.
- **Severe Disease:** Record extensive leaf damage, merging lesions, drying, or premature defoliation.
- **Assessment Note:** The retrieved target leaf spot section of the ICAR–CICR North Zone guide identifies the pathogen and disease cycle, but the available excerpt does not provide a complete numerical disease rating scale for this disease. Therefore, a numerical scale is not reproduced here.

Disease Cycle and Epidemiology

- **Survival in Crop Debris:** The pathogen can survive in infected crop residues, which may act as a source of primary inoculum.
- **Infection and Spread:** Fungal spores can spread from infected material to healthy leaves under suitable environmental conditions.
- **Repeated Infection Cycles:** Target leaf spot can develop through repeated infection cycles when weather conditions remain favourable.
- **Favourable Weather:** Moisture and humid conditions can favour the development and spread of foliar fungal diseases, including target leaf spot.
- **Canopy Conditions:** Dense foliage can retain moisture and reduce air movement within the crop canopy, potentially favouring disease development.
- **Disease Persistence:** Infected plant material left in the field may contribute to disease carryover into subsequent cropping periods.

Favourable Conditions for Disease Development

- Presence of the pathogen in infected crop debris or other infected host material.
- Weather conditions that favour fungal infection and repeated disease cycles.

- Dense crop canopies with reduced air circulation and prolonged leaf wetness.
- Repeated occurrence of the disease in fields where infected plant material remains.
- Delayed identification and management when symptoms begin to spread across the crop.

Prevention and Cultural Management

- **Recommended Varieties:** Grow cotton varieties or hybrids recommended for the region. Use documented resistance or tolerance information where it is available.
- **Field Sanitation:** Manage infected cotton residues appropriately after harvest to reduce potential sources of fungal inoculum.
- **Canopy Management:** Follow locally recommended crop spacing and canopy management practices to support air circulation and reduce prolonged moisture retention.
- **Balanced Crop Management:** Maintain suitable nutrition and irrigation practices to support crop health and avoid unnecessary plant stress.
- **Regular Monitoring:** Inspect leaves throughout crop development, particularly when humid weather or repeated rainfall favours foliar disease development.
- **Avoid Unnecessary Irrigation Wetting:** Where practical, use irrigation methods and schedules that avoid unnecessarily prolonged leaf wetness.
- **Crop Residue Management:** Do not leave heavily infected residues unmanaged. Follow local agricultural recommendations for safe residue handling and disposal.

Biological Management

- **Recommended Biopesticides:** The ICAR–CICR guide recommends seed and soil treatments with suitable biopesticides as part of its broader cotton disease management guidance.
- **Disease-Specific Evidence:** The available target leaf spot excerpt does not identify a particular biological formulation, rate, or application schedule specifically validated for this disease.
- **Product Selection:** Use biological products only when their use is recommended for the crop and intended purpose by local agricultural authorities or the product label.
- **Integrated Approach:** Combine any locally recommended biological treatment with field sanitation, monitoring, and appropriate crop management rather than relying on a single intervention.

Chemical Management

- **Fungicide Recommendation:** The ICAR–CICR guide recommends the use of recommended, appropriately timed fungicides as part of cotton disease management.
- **Correct Identification:** Confirm that the symptoms are consistent with target leaf spot before selecting a fungicide, since different leaf spot diseases may appear similar.

- **Approved Products:** Use only fungicides currently registered and labelled for the intended use in cotton in the relevant region.
- **Application Timing:** Follow current local recommendations for application timing and repeat treatment only as directed by the approved label.
- **Resistance Management:** Where fungicides are recommended, follow label directions on mode-of-action rotation and restrictions to reduce the risk of fungicide resistance.
- **Safe Use:** Wear the required protective equipment, follow all label precautions, and prevent spray drift or contamination of water sources.
- **Local Validation:** Confirm the suitable product and application schedule with the local agricultural extension service or Krishi Vigyan Kendra. The available source excerpt does not specify a verified product rate for target leaf spot.

Integrated Disease Management (IDM)

1. Grow varieties or hybrids recommended for the local cotton-growing region.
2. Inspect the crop regularly for circular lesions with concentric rings and dark centres.
3. Record the location and extent of affected areas and monitor whether new lesions are appearing.
4. Manage infected crop residues after harvest according to local agricultural guidance.
5. Maintain suitable crop spacing and canopy management to support air circulation and reduce prolonged leaf wetness.
6. Use balanced fertilizer and irrigation practices to maintain crop health.
7. Avoid unnecessary pesticide applications and confirm disease identity before treatment.
8. If fungicide treatment is warranted, use a currently approved product according to its label and local recommendations.
9. Record disease observations, weather conditions, crop stage, and management actions to help assess disease development and improve planning for the next season.

Monitoring and Field-Level Decision Making

- **What to Monitor:** Circular leaf lesions, concentric target-like rings, dark lesion centres, enlargement of spots, leaf drying, and premature defoliation.
- **When to Monitor:** Inspect the crop regularly, especially during periods of humid weather or repeated rainfall.
- **Where to Inspect:** Examine leaves from different parts of the field and check both early symptoms and areas where leaf damage is increasing.
- **Disease Progress:** Record whether lesions remain isolated, become more numerous, merge, or cause substantial leaf damage.

- **Treatment Decision:** Consider disease severity, crop stage, weather, and local recommendations before deciding whether a fungicide application is justified.
- **Seek Confirmation:** Consult an agricultural specialist or plant pathology laboratory if symptoms are unusual or difficult to distinguish from other leaf spot diseases.

Effects on Crop Growth and Yield

- Leaf infection reduces the healthy photosynthetic area available to the plant.
- Severe leaf damage may cause premature defoliation and reduce the plant's ability to sustain normal growth.
- Disease-related loss of foliage may affect crop performance, particularly when substantial infection occurs during important growth stages.
- The actual effect on yield depends on disease severity, crop stage, weather conditions, cotton variety or hybrid, and the effectiveness of management practices. A single yield-loss percentage should not be assumed for every field.

Important Information for Farmers

- Target leaf spot is caused by *Corynespora cassiicola*, a fungal pathogen with a wide host range.
- The target-like pattern of concentric rings is an important symptom, but other causes of leaf spots should be considered during diagnosis.
- Regular monitoring, appropriate canopy management, and management of infected crop residues are useful components of disease prevention.
- Use fungicides only when justified by diagnosis and current local recommendations. Do not assume that a product recommended for another cotton disease will necessarily control target leaf spot.
- The ICAR–CICR North Zone guide recommends resistant varieties or hybrids where available, continued disease monitoring, and recommended fungicide use.
- Confirm the suitability of all treatments with local agricultural authorities before implementation, particularly when applying recommendations outside the region covered by the cited guide.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, 2026. Target leaf spot of cotton identification and disease management section, pp. 111–112.
2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Cotton disease identification and integrated disease management guidance.

1.7 Alternaria Leaf Spot of Cotton

Disease Name and Causal Organism

Alternaria leaf spot is a fungal disease of cotton caused by three species:

- *Alternaria macrospora* Zimm.
- *Alternaria alternata* (Fr.) Keissler.
- *Alternaria gossypina*.

Among these, *A. macrospora* is reported as the major causal species, followed by *A. alternata*. Their association can lead to blight of cotton. The disease occurs in cotton-growing areas and may become particularly severe under favourable weather conditions.

Symptoms and Identification

Symptoms vary depending on the *Alternaria* species involved and the prevailing environmental conditions.

- **Initial symptoms:** Small, circular, brown lesions approximately 0.5–2.5 mm in diameter, often surrounded by a purple margin.
- **Symptoms caused by *A. macrospora*:** Brown, greyish-brown or tan lesions develop mainly on lower leaves. Lesions may enlarge to approximately 3–10 mm in diameter and develop concentric rings with a cracked centre.
- **Symptoms caused by *A. alternata*:** Purple lesions with purple margins develop. Severe infection can cause foliage to dry and leaves to fall prematurely.
- **Symptoms caused by *A. gossypina*:** Yellowing and drying of leaves, premature defoliation and boll rotting may occur.
- **Advanced symptoms:** Extensive necrotic spotting, rapid defoliation and abundant fungal sporulation on fallen leaves may occur under favourable weather conditions.
- **Plant parts most affected:** Older, lower leaves are generally more severely affected.
- **Crop susceptibility:** Cotton plants carrying a heavy boll load may be more susceptible to severe disease.

Disease Rating and Field Assessment

The ICAR–CICR North Zone guide provides the following incidence-based disease rating scale. The scale should be interpreted as the proportion of plants showing infection, rather than the percentage of leaf area affected.

- **Scale 0:** All plants in the field or entry are free from infection.
- **Scale 1:** Approximately 5% disease incidence.
- **Scale 2:** Approximately 6–15% disease incidence.
- **Scale 3:** Approximately 16–25% disease incidence.
- **Scale 4:** More than 26% disease incidence.

For field surveillance, inspect plants across representative parts of the field and record the number showing characteristic leaf spots. Calculate disease incidence as follows:

$$Diseaseincidence(\%) = \frac{Numberofinfectedplants}{Totalnumberofplantsobserved} \times 100$$

The incidence-based rating scale above should not be confused with a Percent Disease Index (PDI) calculated from a leaf-severity scale.

Disease Cycle and Epidemiology

Alternaria leaf spot is a polycyclic disease, meaning that the pathogen can complete multiple infection cycles during a single cropping season when conditions are favourable.

- **Primary inoculum:** Infected crop debris and seed are reported as important sources of primary inoculum.
- **Survival:** Fungal spores can survive adverse environmental conditions and initiate infection when suitable conditions return.
- **Repeated infection:** Newly produced spores can cause additional infections, allowing the disease to spread and intensify during the season.
- **Leaf-age effect:** Sporulation is greater on older leaves than on younger leaves under the conditions described in the source guide.
- **Seasonal development:** Disease incidence is generally low during early vegetative growth, increases around squaring and flowering, and may become most severe during boll development, maturity and boll opening.

Favourable Conditions

The following environmental conditions favour disease development:

- Temperature of approximately 20–28°C, particularly when accompanied by frequent rainfall.
- Relative humidity of approximately 65–90%.
- Leaf-wetness periods sufficiently long for infection, especially within the reported favourable temperature range of 20–30°C.
- Optimum temperature for sporulation of approximately 25–30°C.
- Dense crop foliage and persistent moisture, which can favour disease development and secondary infection.
- Plants with a heavy boll load, which may be more susceptible to severe symptoms.

The source guide reports that sporulation can occur at temperatures up to approximately 35°C. Temperature alone does not determine disease severity; moisture, leaf wetness, inoculum availability and crop condition also matter.

Prevention and Cultural Management

- **Resistant or tolerant cultivars:** Prefer recommended varieties or hybrids with resistance or tolerance where suitable options are available for the region.
- **Crop-residue management:** Remove and appropriately manage infected crop residues after harvest to reduce potential sources of fungal inoculum.
- **Field sanitation:** Maintain field hygiene and manage volunteer cotton plants and infected plant material where relevant.
- **Seed health:** Use healthy seed from reliable sources and follow officially recommended seed-treatment practices.
- **Crop canopy:** Follow recommended plant spacing and agronomic practices to avoid unnecessarily dense foliage and prolonged moisture retention.
- **Irrigation management:** Avoid unnecessary prolonged leaf wetness where irrigation method and farm conditions permit.
- **Seasonal planning:** Follow locally recommended sowing dates and crop-duration guidance to reduce the likelihood of highly susceptible crop stages coinciding with persistently wet weather.

Biological Management

- Use only recommended biological seed or soil treatments where these are registered or officially advised for cotton disease management.
- Apply biological products according to their label directions and the locally recommended application method.
- Treat biological products as components of integrated disease management rather than as guaranteed stand-alone cures for an established foliar infection.

The cited North Zone guide recommends seed and soil treatment with recommended biopesticides and chemicals but does not establish a universally applicable biological product or dose for every cotton-growing region.

Chemical Management

- Consider fungicide application when disease is confirmed and the crop condition, weather and locally recommended disease-management guidance justify treatment.
- Use only fungicides currently registered and officially recommended for Alternaria leaf spot on cotton in the relevant jurisdiction.
- Follow the product label for dose, dilution, application interval, protective equipment, pre-harvest interval and other safety requirements.
- Rotate fungicides with different modes of action when advised by current resistance-management guidance.
- Avoid unnecessary or repeated sprays without field assessment, as this can increase cost and selection pressure for fungicide resistance.

No specific fungicide product, dose or spray schedule is prescribed here because these details must be verified against current registration and regional recommendations.

Integrated Disease Management

An integrated approach combines prevention, monitoring and appropriately selected interventions.

1. Select recommended resistant or tolerant varieties or hybrids where available.
2. Begin field inspection during vegetative growth and increase monitoring as the crop approaches squaring, flowering and boll development.
3. Identify characteristic lesions, paying particular attention to older leaves in the lower canopy.
4. Record disease incidence consistently and compare observations over time to determine whether disease is increasing.
5. Manage infected residues and follow recommended seed-health and field-sanitation practices.
6. Use recommended biological or chemical treatments only where suitable and according to current official guidance.
7. Reassess the crop after any intervention and continue monitoring, particularly during humid or rainy weather.

Monitoring and Field-Level Decision Making

- Inspect representative sections of the field rather than relying on a single affected plant or field edge.
- Record the presence of brown or purple leaf spots, concentric rings, yellowing, necrosis and premature leaf fall.
- Examine older lower leaves carefully, while also checking the rest of the canopy for disease spread.
- Calculate disease incidence using the number of infected plants and the total number inspected.
- Repeat assessments to establish whether the disease is stable or progressing.
- Give additional attention to fields with persistent humidity, frequent rainfall, prolonged leaf wetness or heavy boll load.
- Seek confirmation from a qualified agricultural extension officer or plant pathology laboratory when symptoms are atypical or could be confused with other leaf diseases.

Effects on Crop Growth and Yield

Severe Alternaria leaf spot can cause extensive leaf necrosis and premature defoliation. Loss of healthy foliage may reduce the plant's capacity to maintain normal crop growth and support boll development. Infection associated with *A. gossypina* may also involve boll rotting. The magnitude of yield or fibre-quality loss depends on disease severity, crop stage, cultivar, weather and the duration of infection.

Important Information for Farmers

- Alternaria leaf spot is caused by fungal pathogens, and symptoms can differ among the species involved.
- Brown or purple lesions, concentric rings and premature defoliation are useful diagnostic clues, but visual symptoms alone may not always confirm the causal species.
- Older lower leaves and plants carrying a heavy boll load deserve particular attention during field inspection.
- Warm conditions combined with high humidity, frequent rainfall and prolonged leaf wetness can favour disease development.
- Early recognition and consistent monitoring help identify disease progression before severe defoliation occurs.
- Disease ratings based on the proportion of infected plants should be kept separate from ratings based on the percentage of leaf area affected.
- Use regionally recommended varieties and follow current official plant-protection advice. Do not apply fungicides without checking registration, label directions and local suitability.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, Regional Station, Sirsa, 2026. Alternaria leaf spot of cotton, pp. 113–115.

2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Section on Alternaria leaf spot, including symptoms, seasonal incidence, epidemiology, surveillance and management.

Regional applicability: The North Zone guide provides the disease description and rating scale used here. Management choices should be checked against current recommendations for the farmer's region before being used in farmer-facing advice.

1.8 Paramyrothecium (Myrothecium) Leaf Spot of Cotton

Disease Name and Causal Organism

Paramyrothecium leaf spot, also known as Myrothecium leaf spot or Myrothecium leaf blight, is a fungal disease of cotton caused by *Paramyrothecium roridum*, formerly known as *Myrothecium roridum*.

The pathogen can cause severe leaf spot and blight in cotton (*Gossypium hirsutum*). Under favourable conditions, infection may also affect seedlings, petioles, stems and bolls.

Symptoms and Identification

Symptoms can occur on leaves and other plant parts and may become severe under humid conditions.

- **Initial symptoms:** Minute, water-soaked, dark-purple spots or streaks appear on leaves.
- **Leaf lesions:** Circular or irregular spots, approximately 2–10 mm in size, develop with light-brown or greyish centres and dark-purple margins.
- **Lesion expansion:** Small lesions may merge into larger necrotic patches that are light brown or grey.
- **Leaf-margin infection:** Lesions may begin near leaf margins and progress inward, producing extensive necrosis.
- **Fungal growth:** Under high humidity, black, cushion-like spore-producing structures called sporodochia may appear on lesions, sometimes with white tufts.
- **Shot-hole symptoms:** The centre of older lesions may dry and fall out, leaving holes in the leaf tissue.
- **Defoliation:** Severe infection can cause premature leaf fall and substantial loss of foliage.
- **Seedling symptoms:** Infected seedlings may develop stem and petiole necrosis.
- **Boll infection:** Bolls may also be affected, particularly under conditions conducive to disease development.
- **Crop susceptibility:** Cotton plants carrying a heavy boll load may be more susceptible to severe infection.

The combination of grey or light-brown necrotic centres, dark margins, black sporodochia under humid conditions and possible shot-hole formation can help distinguish this disease from other cotton leaf spots. Visual diagnosis should be confirmed where symptoms are unclear.

Disease Rating and Field Assessment

The ICAR–CICR North Zone guide provides the following incidence-based rating scale:

- **Scale 0:** All plants in the field or entry are free from infection.
- **Scale 1:** Approximately 5% disease incidence.
- **Scale 2:** Approximately 6–15% disease incidence.
- **Scale 3:** Approximately 16–25% disease incidence.
- **Scale 4:** More than 26% disease incidence.

Calculate disease incidence using the number of infected plants observed in a representative field sample:

$$Diseaseincidence(\%) = \frac{Numberofinfectedplants}{Totalnumberofplantsobserved} \times 100$$

This scale describes the proportion of plants infected. It is not a leaf-area severity scale or a Percent Disease Index (PDI) scale. The source does not provide a separate numerical PDI calculation for this disease in the cited section.

Disease Cycle and Epidemiology

The fungus survives between cropping seasons and spreads when environmental conditions favour infection.

- **Off-season survival:** The pathogen survives as spores (conidia) or mycelium on infected crop debris left in the field and may occasionally survive on seed.
- **Primary infection:** Surviving fungal material can provide inoculum for infection in the following crop.
- **Spore dispersal:** During rainy periods, spores can reach healthy leaves through rain splash, irrigation water and wind.
- **Infection:** Spores germinate and penetrate leaf tissue when moisture is present on the leaf and humidity is high.
- **Secondary spread:** Black, cushion-like sporodochia form within lesions and produce spores that can spread to healthy leaves and plants through wind and water.
- **Repeated infection:** The pathogen can produce successive infection cycles during favourable periods.
- **Return to survival phase:** Necrotic tissue may dry and fall to the soil, allowing the fungus to persist on plant debris.

Favourable Conditions

Conditions that favour disease development include:

- High relative humidity, reported at around 85% in the disease-cycle description.
- A film of water on leaf surfaces, particularly during rainy or humid weather.
- Frequent rainfall or irrigation that facilitates spore movement and prolongs leaf wetness.
- Infected crop residues or contaminated seed that provide a source of inoculum.
- Dense foliage or a crop canopy that retains moisture and restricts air movement.
- Cotton plants carrying a heavy boll load, which may be more susceptible to severe symptoms.

The cited North Zone guide does not specify a complete numerical temperature range for this disease in the retrieved section. Temperature-based thresholds should therefore not be inferred from the conditions listed above.

Prevention and Cultural Management

- **Resistant or tolerant cultivars:** Prefer recommended cotton varieties or hybrids with resistance or tolerance where suitable options are available.
- **Crop-residue management:** Remove and appropriately manage infected crop residues after harvest to reduce potential sources of fungal inoculum.
- **Field sanitation:** Keep the field free of infected plant debris where practicable and manage volunteer cotton plants that may contribute to disease carryover.
- **Seed health:** Use healthy seed from reliable sources and follow officially recommended seed-treatment practices.
- **Canopy management:** Follow recommended spacing and agronomic practices to avoid unnecessarily dense foliage and prolonged moisture retention.
- **Irrigation management:** Where feasible, avoid irrigation practices that unnecessarily prolong leaf wetness.
- **Crop monitoring:** Inspect fields regularly, particularly during humid and rainy periods, to detect early symptoms before severe leaf blight and defoliation develop.

Biological Management

- The ICAR–CICR North Zone guide recommends seed and soil treatment using recommended biopesticides and chemicals as part of disease management.
- Use only biological products recommended for the intended treatment and follow their label directions.
- Apply biological treatments as components of an integrated programme rather than assuming they will cure severe established infections.

The cited North Zone section does not specify a particular biological product, application rate or schedule for Paramyrothecium leaf spot. Such details should be obtained from current regional recommendations.

Chemical Management

- Consider fungicide treatment when the disease is identified and current local recommendations support intervention.
- Use only fungicides currently registered and recommended for the intended use on cotton in the relevant jurisdiction.
- Follow the product label for dose, dilution, application interval, protective equipment and pre-harvest requirements.
- Avoid repeated or unnecessary fungicide applications and follow current resistance-management guidance.
- Continue field monitoring after treatment to assess whether disease progression has slowed.

The cited North Zone guide recommends the use of recommended, appropriately timed fungicides but does not specify a product or dose in the retrieved disease section. A particular chemical treatment should not be inferred from that general recommendation.

Integrated Disease Management

An integrated management programme should combine prevention, field surveillance and suitable interventions.

1. Select recommended resistant or tolerant cotton varieties or hybrids where available.
2. Use healthy seed and follow locally recommended seed-treatment practices.
3. Manage infected crop debris and maintain field sanitation to reduce potential sources of inoculum.
4. Inspect representative areas of the field, paying attention to leaf margins, lower foliage and plants showing premature defoliation.
5. Look for characteristic necrotic lesions, dark margins, black sporodochia and shot-hole symptoms.
6. Record disease incidence and repeat assessments to identify whether infection is increasing.
7. Apply recommended biological or chemical treatments only when appropriate and according to current official guidance.
8. Reassess the crop after intervention and continue monitoring during periods of high humidity and rainfall.

Monitoring and Field-Level Decision Making

- Examine plants across representative parts of the field rather than relying only on field-edge observations.
- Inspect leaves for water-soaked purple spots, grey or light-brown necrotic centres, dark margins and expanding lesions.
- Check for black spore-producing structures and holes where the centres of older lesions have fallen out.
- Examine seedlings, petioles, stems and bolls if symptoms extend beyond the leaf surface.
- Calculate disease incidence using a consistent sampling method and record observations over time.
- Increase surveillance during humid weather, frequent rainfall or periods of prolonged leaf wetness.
- Seek assistance from an agricultural extension officer or plant pathology laboratory when symptoms overlap with those of other fungal diseases.

The cited source provides an incidence-based rating scale but does not establish a separate economic threshold for this disease in the retrieved North Zone section. Management decisions should therefore consider disease progression, crop stage, local advisories and current recommendations rather than relying on an unsupported threshold.

Effects on Crop Growth and Yield

Severe Paramyrothecium leaf spot can cause extensive leaf necrosis and premature defoliation, reducing the amount of healthy foliage available to support crop growth and boll development. Infection may also affect seedlings and bolls. The extent of crop damage depends on disease severity, crop stage, cultivar susceptibility, weather conditions and the duration of infection.

Important Information for Farmers

- Paramyrothecium leaf spot is a fungal disease that can cause both leaf spot and blight.
- Circular or irregular necrotic spots with dark margins and grey or light-brown centres are important identification clues.
- Black sporodochia under humid conditions and shot-hole formation in older lesions can provide additional diagnostic evidence.
- Infected crop residues and, occasionally, seed can carry the pathogen between seasons.
- Rain splash, irrigation and wind can contribute to the spread of spores.
- High humidity and leaf wetness favour infection, so regular field inspection is particularly important during wet weather.
- Use the incidence rating scale consistently and do not confuse the percentage of infected plants with the percentage of leaf area damaged.
- Confirm current regional recommendations before selecting any fungicide or biological treatment.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, Regional Station, Sirsa, 2026. Paramyrothecium/Myrothecium leaf spot of cotton, pp. 115–116.

2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Section on Myrothecium leaf spot, including disease symptoms, seasonal incidence and epidemiology.

Regional applicability: The disease description and rating scale follow the cited ICAR–CICR North Zone guide. Management recommendations should be checked against current official guidance for the farmer's region before being used in farmer-facing advice.

1.9 Cotton Boll Rot Complex (Internal and External Boll Rot)

Disease Name and Causal Organism

Cotton boll rot complex refers to rotting of cotton bolls caused by different pathogens that infect developing or mature bolls. The North Zone guide discusses boll rot as a complex involving internal and external boll rotting. The integrated management guide identifies several fungal species as contributors to fungal boll rot, while bacterial internal boll rot may also occur.

Because boll rot can have multiple causes, the causal organism should not be assigned from boll appearance alone.

Symptoms and Identification

Symptoms vary according to the pathogen involved, the stage of boll development and environmental conditions.

- **External boll rot:** Visible discolouration, lesions or rotting develops on the boll surface.
- **Internal boll rot:** The internal tissues of the boll may rot, even when external symptoms are limited or not immediately obvious.
- **Damaged locules:** Infected portions of the boll may contain discoloured, deteriorated or rotting tissues.
- **Seed and lint damage:** Infection can reduce the quality of seed and lint within affected bolls.
- **Advanced infection:** Severely affected bolls may deteriorate and become unsuitable for normal harvesting or use.
- **Variable symptoms:** The appearance of boll rot differs among pathogens; symptoms may overlap with damage caused by insects, weather or other causes.

The sources describe boll rot as a complex rather than a single disease with one universal symptom pattern. Laboratory or expert diagnosis may be required to distinguish bacterial, fungal and other causes.

Disease Rating and Field Assessment

The retrieved sections of the ICAR–CICR North Zone guide do not provide a complete disease-specific numerical rating scale for the cotton boll rot complex. Therefore, a scale should not be invented or substituted from another cotton disease.

For practical field assessment, record the following observations:

- Number of plants examined and number of plants bearing affected bolls.
- Number of affected bolls and total bolls examined.
- Whether symptoms are mainly external, internal or both.
- Crop stage and the distribution of affected bolls within the field.
- Recent rainfall, humidity, irrigation and other conditions that may be associated with boll deterioration.

- Presence of insect injury or other visible damage that may complicate diagnosis.

Where a numerical incidence estimate is required, calculate boll incidence using a clearly defined sampling method:

$$Affectedbollincidence(\%) = \frac{Numberofaffectedbolls}{Totalnumberofbollsexamined} \times 100$$

This calculation is a general field-recording method and is not presented as an official disease rating scale from the cited guide.

Disease Cycle and Epidemiology

The boll rot complex can involve different pathogens and infection routes. The precise disease cycle depends on the causal organism.

- **Multiple causal organisms:** Different fungal and bacterial pathogens may be associated with boll rotting.
- **Infection of bolls:** Pathogens can affect developing or mature bolls under suitable conditions.
- **Weather influence:** Rainfall, high humidity and prolonged moisture can favour some boll-rotting pathogens.
- **Plant injury:** Insect damage or other physical injury may provide opportunities for some pathogens to enter boll tissues.
- **Crop residues:** Depending on the pathogen, infected plant material may contribute to pathogen survival between seasons.

These points describe general features of boll rot complexes. The retrieved source material does not establish one complete disease cycle applicable to every pathogen associated with cotton boll rot.

Favourable Conditions

Conditions that may favour boll rotting include:

- Frequent rainfall and prolonged wet conditions during boll development and maturation.
- High humidity and moisture retained around bolls.
- Dense crop canopies that dry slowly after rain or irrigation.
- Insect injury or other damage to boll tissues.
- Presence of susceptible cotton plants and pathogen inoculum in the crop environment.

The importance of each factor depends on the causal organism. The cited material does not establish a single numerical temperature or humidity threshold for the entire boll rot complex.

Prevention and Cultural Management

- **Field monitoring:** Inspect developing and mature bolls regularly, particularly after periods of persistent rainfall or high humidity.
- **Crop sanitation:** Manage infected plant material appropriately after harvest, following local crop-residue recommendations.
- **Canopy management:** Follow recommended plant spacing and agronomic practices to avoid unnecessarily dense growth and prolonged moisture retention.
- **Insect management:** Monitor boll-feeding insects and manage them according to current integrated pest management recommendations to reduce avoidable boll injury.
- **Harvest management:** Harvest mature, healthy open bolls promptly and keep harvested cotton protected from rain and excess moisture.
- **Diagnosis:** Seek expert confirmation when boll symptoms are unclear or when bacterial, fungal and insect-related damage may be confused.

Biological Management

- Use biological products only where they are recommended for the diagnosed pathogen and intended use in cotton.
- Follow the product label and current regional agricultural recommendations.
- Do not assume that a biological product effective against one fungal disease will control every cause of boll rot.

The retrieved source sections do not provide a complete, disease-specific biological treatment schedule for the entire boll rot complex. A particular product or dose should not be inferred.

Chemical Management

- Select chemical treatments only after considering the likely cause of boll rot and current regional recommendations.
- Use only products registered for the intended crop, pathogen and application method.
- Follow the product label for dose, application timing, protective equipment, preharvest interval and other safety requirements.
- Avoid indiscriminate fungicide use where the cause has not been identified, particularly when symptoms may be bacterial or associated with insect injury.
- Continue monitoring after any intervention to assess whether new boll infections are occurring.

The sources used here do not provide one universally applicable chemical treatment for all internal and external boll rots. A product-specific recommendation requires confirmation of the causal organism and current official guidance.

Integrated Disease Management

1. Inspect the crop regularly from boll development through maturation.
2. Record affected bolls and distinguish visible external rot from suspected internal rot where possible.
3. Note weather conditions, crop stage and any insect injury associated with affected bolls.
4. Manage boll-feeding insect pests according to current integrated pest management guidance.
5. Maintain appropriate crop spacing and agronomic practices to reduce unnecessarily prolonged moisture retention.
6. Manage infected crop residues and harvested cotton appropriately.
7. Obtain diagnostic assistance where the causal organism is uncertain.
8. Use biological or chemical interventions only when supported by diagnosis and current regional recommendations.

Monitoring and Field-Level Decision Making

- Examine representative plants from different parts of the field.
- Inspect bolls at different stages of development and maturity.
- Record the number of affected bolls, their symptoms and their distribution in the field.
- Check whether affected bolls show insect injury, external lesions or internal deterioration.
- Increase surveillance after prolonged wet weather or when boll rot symptoms begin to appear.
- Separate confirmed disease observations from suspected cases in field records.
- Seek expert diagnosis before making pathogen-specific treatment decisions.

No universal economic threshold for the entire cotton boll rot complex is established in the retrieved source sections. Decisions should be based on diagnosis, severity, crop stage and current regional recommendations.

Effects on Crop Growth, Yield and Quality

Boll rot can reduce the number of healthy harvestable bolls and damage seed and lint quality. Depending on the causal organism and severity, affected bolls may deteriorate before picking, resulting in yield loss and reduced marketable cotton quality. The magnitude of loss depends on pathogen identity, weather, crop stage and the proportion of bolls affected.

Important Information for Farmers

- Cotton boll rot is a disease complex and may involve different fungal or bacterial pathogens.
- Internal boll rot and external boll rot should not automatically be assumed to have the same cause.
- Rainfall, humidity, prolonged moisture and boll injury can contribute to disease development, depending on the pathogen.
- Regular inspection during boll development and maturation helps identify affected bolls early.
- Record the number of affected bolls consistently, but do not treat a general incidence calculation as an official disease rating scale.
- Fungicide treatment should not be recommended indiscriminately for all boll rots; diagnosis and current local guidance are important.
- Protect harvested cotton from excess moisture and follow recommended harvesting and post-harvest practices.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, Regional Station, Sirsa, 2026. Disease identification and monitoring section covering cotton diseases, pp. 102–123.
2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Sections on cotton disease importance and fungal boll rot disease.

Source limitation: The cited material identifies fungal boll rot as involving several fungal species and discusses internal and external boll rot within the cotton disease-management context. The retrieved excerpts do not provide a complete causal-organism list, a unified disease cycle, or a specific rating scale and treatment schedule for all forms of boll rot. These details have therefore not been invented.

1.10 Nematode Problem in Cotton

Disease Name and Causal Organisms

Nematode infestation is an important below-ground problem in cotton. The ICAR–Central Institute for Cotton Research identifies two economically important plant-parasitic nematodes:

- **Root-knot nematode:** *Meloidogyne incognita*, reported as dominant in the northern cotton-growing areas of India.
- **Reniform nematode:** *Rotylenchulus reniformis*, reported to infest cotton in central and southern India.

Nematodes can also participate in disease complexes, increasing the incidence and severity of wilt diseases and post-emergence damping-off caused by pathogenic fungi.

Symptoms

- **Root galls:** Root-knot nematode infection produces characteristic swellings or knots on the roots.
- **Reduced root system:** Infected plants develop smaller root systems with fewer lateral and feeder roots.
- **Plant stunting:** Plants remain short and show poor growth.
- **Leaf chlorosis:** Leaves may become pale or yellow due to impaired water and nutrient uptake.
- **Wilting:** Infected plants may wilt early; mature plants may show temporary wilting during afternoons.
- **Uneven crop growth:** Infested fields may contain irregular patches of pale, stunted and unhealthy plants resembling nutrient or water deficiency.
- **Seedling mortality:** A high nematode population at sowing may kill plants during the seedling stage.

Disease Cycle and Epidemiology

- The root-knot nematode life cycle typically takes approximately 4–6 weeks, depending on soil temperature.
- Mature females lay egg masses in a protective, jelly-like matrix on the root surface or within root galls.
- The first juvenile stage develops inside the egg. The second-stage juvenile (J2) hatches and is the mobile stage that moves through soil water to locate and penetrate young cotton roots.
- After entering the root, the J2 establishes a permanent feeding site and induces the formation of enlarged feeding cells, resulting in visible root galls.
- The nematode becomes sedentary and develops through subsequent juvenile stages before reaching adulthood.
- Nematode damage can interact with soil-borne pathogenic fungi, increasing wilt and post-emergence damping-off.
- The ICAR–CICR integrated management guide identifies sandy soils as favourable for multiplication of root-knot nematodes in cotton.

Effects on Crop Growth and Yield

- Reduced root function restricts the uptake of water and essential nutrients.
- Plants may produce fewer and smaller bolls.
- Severe infestation can reduce lint weight and adversely affect fibre quality, including staple length and strength.
- The ICAR–CICR integrated management guide reports yield losses of approximately 10–50% under moderate to severe infestation.

Disease Rating Scale

The retrieved section of the ICAR–CICR North Zone guide does not provide a numerical disease rating scale or a specific economic threshold for cotton nematode infestation. Therefore, no numerical severity grades are assigned here.

Monitoring and Early Detection

- Inspect uneven or stunted patches where plants appear pale, unhealthy or wilted despite apparently adequate crop care.
- Examine roots of affected plants for characteristic galls and reduced development of lateral and feeder roots.
- Consider nematode infestation when root symptoms occur alongside unexplained stunting, chlorosis or wilting.
- Because above-ground symptoms may resemble water stress or nutrient deficiency, confirm suspected infestation through appropriate root and soil examination rather than relying on foliage symptoms alone.

Management

The retrieved North Zone guide section identifies the principal nematode species, symptoms and disease interactions but does not provide a complete, nematode-specific management schedule. Accordingly, the following source-based considerations should guide management planning:

- Identify the nematode involved before selecting a management intervention.
- Assess the root system and field history when nematode infestation is suspected.
- Account for the interaction between nematodes and soil-borne fungal pathogens when investigating wilt and post-emergence damping-off.
- Select any nematode-management treatment according to current ICAR–CICR or State Agricultural University recommendations and the applicable product label.

Other Relevant Information

- Nematodes are often described as a hidden enemy because much of their damage occurs below the soil surface.
- Root-knot nematode is the principal species highlighted for northern cotton-growing areas in the North Zone guide.
- Reniform nematode is highlighted for central and southern cotton-growing regions.
- Root galling, stunting and chlorosis are useful warning signs, but diagnosis should distinguish nematode damage from other causes of poor crop growth.

Sources Cited

1. *An Illustrative Guide to Cotton Cultivation in the North Zone of India.* ICAR–Central Institute for Cotton Research, Regional Station, Sirsa, 2026. Section on "Nematodes problem in Cotton," pp. 121–122.

2. *Integrated Management of Cotton Pests, Diseases and Nematodes.* ICAR–Central Institute for Cotton Research. Section on plant-parasitic nematodes.

Source note: Numerical yield-loss estimates and the additional life-cycle details are taken from the ICAR–CICR integrated management guide. A nematode-specific rating scale and complete treatment schedule were not present in the retrieved North Zone guide section.